

Tampere University
Unit of Computing Sciences
COMP.SE.610 Software Engineering Project 1
COMP.SE.620 Software Engineering Project 2

Group name/number: Theeta-G26

Project name: North star Backend and Watch integration

Project plan

Abdul Wasay Rizwani
Nauman Afzal
Ibtehaj Abbas
Waqas Hameed
Irteza Hammad
Sumair ul Hasan

Version history

Version	Date	Author	Description
0.1	15.10.2013	Abdul Wasay	First draft

Table of Contents

1. INTRODUCTION	5
1.1 PURPOSE AND SCOPE OF PROJECT	5
1.2 PRODUCT AND ENVIRONMENT	5
1.3 CUSTOMER'S CURRENT SYSTEM AND OTHER SIMILAR SYSTEMS.....	6
1.4 PROJECT CONSTRAINTS.....	6
1.4.1 Development Constraints.....	6
1.4.2 Quality and Security Constraints.....	6
1.4.3 Legal Constraints (IPR & GDPR).....	6
1.4.4 Ethical and Sustainability Constraints	7
1.5 DEFINITIONS, ABBREVIATIONS AND ACRONYMS	7
2. PROJECT ORGANISATION	9
2.1 GROUP MEMBERS.....	9
2.2 CUSTOMER	10
2.3 RELATED ORGANISATIONS	10
3. PROJECT GOALS AND ENDING/TERMINATION	11
3.1 HIGH LEVEL GOALS OF THE PROJECT GROUP	11
3.2 HIGH LEVEL GOALS OF THE CUSTOMER	11
3.3 HIGH LEVEL GOALS AND DELIVERABLE(S) OF THE PROJECT	11
3.4 QUITTING (TERMINATION) CRITERIA OF THE PROJECT	12
3.5 ENDING CRITERIA OF THE PROJECT	12
4 PROJECT AND PROCESS MANAGEMENT	13
4.1 METHODS AND TOOLS	13
4.1.1 Communication and Collaboration Tools.....	13
4.1.2 Demo Deployment.....	13
4.1.3 Project Documentation.....	13
4.2 PLANNED MEETINGS	14
4.3 LEARNING AND STUDY PLAN	14
5 PROJECT ITERATIONS AND TIMING.....	15
5.1 ITERATIONS (SPRINTS).....	15
5.1.1 Iteration 1 (First Implementation Sprint).....	16
6 REQUIREMENTS.....	16
6.1 FUNCTIONAL REQUIREMENTS.....	16
6.2 NON-FUNCTIONAL REQUIREMENTS GOALS.....	17
6.2.1 Usability goals.....	17
6.2.2 Performance goals.....	17
6.2.3 Reliability goals.....	17
6.2.4 Security goals.....	17
7. RISK MANAGEMENT	18
7.1 RISK LIST.....	18
7.2 RISK MONITORING	19

8 REFERENCES 19

9 OPEN ISSUES 19

10 IDEAS FOR FURTHER DEVELOPMENT 19

APPENDIX A..... 19

APPENDIX B..... 19

1. Introduction

1.1 Purpose and scope of project

Theeta group's fall 2025 Software Engineering Project course. It outlines the project's developmental goals and will be updated as needed throughout the course.

Initially, the project aimed to enhance the existing application and introduce new features, based on the customer's proposal. However, this direction shifted after an external consultancy, Mindhive, assessed the current application and recommended next steps.

Mindhive provided two key documents: one detailing the existing application's state and another, "The North Star", presenting a high-level architectural vision. The assessment revealed the old application's limitations in scalability, maintainability, and overall quality, rendering it unsuitable for its size and complexity. Conversely, The North Star proposed a reimagined, improved architecture, aligned with industry standards and best practices, to guide the development of a new application.

Based on Mindhive's input, the customer opted for a complete rework instead of incremental improvements. The team agreed, and the project scope was defined to follow The North Star vision, prioritise quality, and develop the application as thoroughly as possible to ensure future teams can continue development without technical hurdles. Moreover, in addition to that team will be responsible for planning the smart watch integration with the new backend to make it work.

1.2 Product and Environment

The product, known as Lumisovellus in Finnish and Snowledge in English, is a web and mobile application designed for visitors and staff at Pallastunturi.

The application offers two primary features:

1. **Environmental Monitoring:** Users can access real-time data on snow levels, temperature, and wind conditions at Pallastunturi. Historical data from the preceding two days is also available.
2. **Safety Assistance:** Exclusively through the mobile application, users within the Pallastunturi area can request and receive assistance.

1.3 Customer's current system and other similar systems

The current system operates slowly and is limited to two backends and two databases. This limitation prevents the integration of smartwatches with the software. Additionally, there is a lack of proper documentation.

1.4 Project Constraints

The project operates under specific legal, quality, and administrative requirements, detailed below.

1.4.1 Development Constraints

- **Technology & Methodology:** Adherence to the North Star architecture is mandatory. Development tools include Git/GitHub, Trello (for Kanban), and MMT (for time logging and risk monitoring).
- **Team Hours:** Each team member is required to log between 110-130 working hours.

1.4.2 Quality and Security Constraints

- **Test Coverage:** A minimum of 80% code coverage is required for all newly developed code.
- **Quality Gates:** The Continuous Integration (CI) pipeline must enforce linting, type-checking, and comprehensive test execution.
- **Security:** Production environments mandate HTTPS and HSTS, robust input validation, rate limiting, and a basic Web Application Firewall (WAF) implementation.

1.4.3 Legal Constraints (IPR & GDPR)

- **Intellectual Property Rights (IPR):** The customer is granted commercial use rights. Group members retain non-commercial rights and ownership of any breakthrough algorithms developed.
- **General Data Protection Regulation (GDPR):** The application may collect precise GPS location data within Finland, personal identifiers (name, email, phone), and technical identifiers (IP address, UDID).
- **GDPR Requirements:** Implementation of parental consent, data retrieval mechanisms, data deletion functionalities, and mapping of data processing to a lawful basis are essential.

1.4.4 Ethical and Sustainability Constraints

- **Sustainable Development Goals (SDGs):** The project implicitly contributes to SDG 3 (Good Health and Well-being) through its safety features and SDG 9 (Industry, Innovation, and Infrastructure) by establishing a maintainable, high-quality, and resilient software infrastructure.
- **Accessibility:** Emphasis on accessibility and intuitiveness is critical to ensure a superior user experience.

1.5 Definitions, abbreviations and acronyms

Term	Definition
CI/CD	Continuous Integration and Continuous Delivery /Deployment. Set of automated software development practices that streamline the process of releasing new features and bug fixes.
Flutter	Framework for creating native apps for mobile, web, desktop, and embedded devices.
HSTS	HTTP Strict Transport Security. Web security standard that forces a web browser to always connect to a website using HTTPS.
HTTPS	Hypertext Transfer Protocol Secure. Encrypts exchanged data between web browser and server.
IDFA	Identifier for Advertisers. A unique random code assigned to a device by Apple. Allows advertising tracking and measurement.
Lumisovellus	Name of the application under development during this project.
Suunto	Name of the watch company.
Mindhive	Software development company, which has consulted the customer, and provided documentation of the current state and proposal of the next steps of the application.

MMT	Metrics Monitoring Tool.
Monorepo	Software development strategy in which the code for multiple projects is stored in the same repository.
MySQL	Database management system.
Next.js	Web development framework.
North Star	Name of the document/instructions describing how the large-scale rewrite could be done for the existing application.
Pallaksen Pöllöt	Project customer. Ski school in Pallastunturi.
PM	Project Manager.
PO	Product Owner.
Snowledge	Name of the application under development during this project in English.
SW	Software.
Trello	Kanban-style list-making application/service.
Turborepo	Build system for JavaScript and TypeScript codebases designed for monorepos.
UDID	Unique Device Identifier. A unique code assigned to a device, e.g. a mobile phone.
UI	User Interface. Here the human-computer interaction and communication happens.
WAF	Web Application Firewall. Filters, monitors and blocks malicious HTTP traffic to and from web applications.

2. Project organization

2.1 Group members

1. Abdul Wasay

- **Email:** wasay.rizwani@tuni.fi
- **Project Role:** Project Manager
- **Experience:** Project management, Backend technologies, mobile development

2. Sumair ul Hasan

- **Email:** sumair.hasan@tuni.fi
- **Project Role:** Senior Developer
- **Experience:** Backend technologies, CI/CD

3. Waqas Hameed

- **Email:** waqas.hameed@tuni.fi
- **Project Role:** Senior Developer
- **Experience:** Backend technologies, Frontend technologies

4. Nauman Afzal

- **Email:** nauman.afzal@tuni.fi
- **Project Role:** Product Owner
- **Experience:** Project management, Requirements Engineering, Backend technologies

5. Irteza Hameed

- **Email:** irteza.hameed@tuni.fi
- **Project Role:** Senior Developer
- **Experience:** Backend technologies, frontend development

6. Ibtehaj Abbas

- **Email:** ibtehaj.abbas@tuni.fi
- **Project Role:** Senior Developer
- **Experience:** Testing technologies, Cloud Technologies

2.2 Customer

Name	Email
Juusto Holstein	juuso@pallaksenpollot.com
Arto Hämäläinen	artohamalainen@gmail.com

2.3 Related organizations

Names	Email	Roke
Tero Ahtee	tero.ahtee@tuni.fi	Project Coach
Pekka Hämäläinen	pekka.makiaho@tuni.fi	Course Staff
Timo Poranen	timo.poranen@tuni.fi	Course Staff
Kari Systä	kari.systa@tuni.fi	Course Staff

3. Project goals and ending/termination

3.1 High level goals of the project group

The new backend has been implemented following the North Star documentation from Mind Hive. A secondary goal is to complete course requirements, achieving a final grade of 4 or higher, while staying on schedule within reasonable working hours (110–130 hours total).

Furthermore, we aim to increase our knowledge of building scalable systems while working in a new team.

3.2 High level goals of the customer

New backend implemented following the North Star documentation from Mind Hive Research: This involves a complete overhaul of the existing backend infrastructure, aligning it with the strategic architectural guidelines and best practices outlined in Mind Hive Research's North Star documentation. This likely includes adopting new technologies, improving scalability, enhancing security, and optimizing performance.

Integrate with Suunto watches (Optional): This outlines a future initiative to establish a connection or compatibility with Suunto smartwatches. This integration could involve data exchange (e.g., fitness metrics, GPS data), enabling new functionalities within the application for Suunto users, or providing a seamless experience across platforms. The "optional" designation suggests it's a desirable but not strictly mandatory feature at this stage.

3.3 High level goals and deliverable(s) of the project

This project focuses on developing a local iteration of the Lumisovellus "North Star" rewrite. Key components include a unified backend, a MySQL database with migrations, and a monorepo (TurboRepo) incorporating pipelines, linting, and testing. A Docker-based containerized environment will be established for local development.

Expected deliverables are a functional prototype showcasing core features, comprehensive architectural and setup documentation, and a fully implemented CI/CD pipeline. Furthermore, a project plan for integrating with Suunto watches will be created, with an optional aim to implement as much of this plan as feasible.

3.4 Quitting (termination) criteria of the project

Project termination may occur due to overwhelming technical difficulties, such as the prototype failing to meet the "North Star" rewrite despite multiple attempts. Serious resourcing issues, like key members quitting and leaving the remaining students unable to continue, can also lead to termination. In either scenario, the decision to terminate will involve the project group, course instructor, and customer.

3.5 Ending criteria of the project

Key Deliverables:

- A local prototype must be delivered by December 12, 2025.
- Accompanying documentation, including test and final reports, must also be submitted by this date.

Customer and Course Requirements:

- The customer will review and accept the prototype.
- The project formally concludes upon completion of the course timeline.
- A final meeting must be held, minutes submitted, and all action points closed from the course's perspective.

Individual Responsibilities:

- Each student is required to complete a minimum of 110 working hours.
- Individual peer reviews must be submitted.

Project Authority:

- The project can be formally ended by the group, the customer, and the course instructor.

4 Project and Process management

4.1 Methods and tools

4.1.1 Communication and Collaboration Tools

- **Internal Communication:** Telegram is the primary channel for urgent internal communications.
- **External Communication:**
 - **Email:** Main method for customer communication.
 - **Telegram Channel:** Facilitates direct communication between customers and team members.
- **Version Control:** Git is used for version control, with project code hosted on GitHub. This switch from GitLab provides direct access to all team members and removes previous member limitations.
- **Project Planning and Management:**
 - **Trello:** A Trello workspace with two Kanban boards (general-purpose and development) visually represents the workflow. Tasks are marked "done" after documentation and passing tests. Trello, in conjunction with the Metrics Monitoring Tool (MMT), supports change management by quickly assessing and integrating changes to prevent development slowdowns.
 - **Metrics Monitoring Tool (MMT):** Used for logging working hours, generating weekly project reports (including statistics like working hours and realized risks), and proactive risk management.
- **Group Meetings:**
 - Google Meet is used for internal group meetings.
 - Microsoft Teams is used for sprint reviews.

4.1.2 Demo Deployment

Demo versions are made available to the customer via the existing domain (e.g., <https://demo.lumisovellus.fi>) or an IP/port combination. The CI/CD pipeline is utilized for deployment.

4.1.3 Project Documentation

Key project documents include:

- Project Plan: Outlines the overall scope.
- Watch integration plan
- Software architecture details and setup documentation.
- Final Report: Presents the overall project outcome.

4.2 Planned Meetings

Meeting Schedule

- **Customer Meetings:** Weekly on Sundays at 6:00 PM.
- **Internal Team Meetings:** Weekly on Saturday evenings at 6:00 PM.
- **Stand-up meetings:** Every Tuesday and Thursday evening at 8:00 PM.

Sprint Meetings (per sprint):

- October 12, 2025 – 6:00 PM
- October 24, 2025 - 6 PM
- November 7, 2025 - 6:00 PM
- November 21, 2025 - 6:00 PM
- December 5, 2025 - 6:00 PM

These sprint meetings will be scheduled by the Project Manager (PM) after one month.

4.3 Learning and study plan

The project group will dedicate significant time to studying CI/CD practices and setting up a robust REST environment. However, the majority of the learning effort will be concentrated on planning and understanding the intricacies of smartwatch integration. This includes researching compatible technologies, data exchange protocols, and user experience considerations for the Lumisovellus application.

5 Project iterations and timing

5.1 Iterations (sprints)

The project is structured into two-week iterations, known as sprints, which serve as incremental steps toward achieving the project's defined scope. While a high-level schedule is provided in Table 5.1, predicting the exact content of each implementation sprint or the final product state at this stage is challenging. The team will prioritize implementing as much of the project as possible, establishing a robust foundation for future development by other teams.

Table 5.1. Project Schedule.

Sprint No.	Schedule (Start/End Date)	High-Level Content
1	(22.09.2025 / 12.10.2025)	Initializing project repository Defining a more detailed product backlog for subsequent sprints
2	(12.10.2025 / 24.10.2025)	- Replicating Api with the new architecture - Starting CI/CD integration
3	(24.10.2025 / 07.11.2025)	- Migrating existing data to the new DB - Starting researching the suunto watch integration
4	(07.11.2025 / 21.11.2025)	- Completing plan for the Suunto watch integration
5	(21.11.2025 / 5.12.2025)	- Defining and scoping feasible work for the final implementation sprint

5.1.1 Iteration 1 (First Implementation Sprint)

Project Initialization

The project begins with "Initializing project repository" to establish a central hub for development. This includes:

- **VCS Setup:** Configure a Version Control System (e.g., Git) for tracking and collaboration.
- **Repository Creation:** Set up the main repository for all project files.
- **Initial Commit:** Add a basic project structure, README, and configuration.
- **Access Control:** Define user permissions for the repository.
- **Tool Integration:** Connect the repository with IDEs, CI/CD, and project management tools.

Product Backlog Refinement

"Defining a more detailed product backlog for subsequent sprints" is an ongoing agile activity, providing a prioritised list of features. This involves:

- **Decomposition:** Break high-level features (epics) into smaller user stories.
- **Acceptance Criteria:** Define conditions for a story's completion.
- **Prioritisation:** Rank stories by business value, dependencies, and risk.
- **Estimation:** Estimate effort for each story (e.g., story points).
- **Stakeholder Collaboration:** Engage with stakeholders for shared understanding.
- **Regular Review:** Continuously update the backlog based on new information and feedback.
- **Readiness for Sprint Planning:** Ensure stories are well understood, estimated, and ready for development.

6 Requirements

6.1 Functional requirements

The system must fulfill the following core functional requirements to achieve the main project goals:

- **Codebase Initialization:** The system shall provide a clean, structured codebase as a foundation for modular development and future feature integration.
- **CI/CD Pipeline Development:** The system shall support automated building, testing, and deployment through a CI/CD pipeline to ensure efficient and error-free code delivery.
- **Unified Database and Backend:** The system shall implement a centralized backend with an integrated database to manage and store all user, device, and activity data consistently.
- **Suunto Watch Integration:** The system shall connect to the Suunto API to retrieve and synchronize fitness and health data from user devices into the platform.

6.2 Non-functional requirements goals

6.2.1 Usability goals

Our product aims for high usability, allowing users to easily navigate and achieve their tasks with minimal effort. The most critical aspects are:

- **Ease of Learning:** New users should be able to understand the basic functionalities within 5 minutes.
- **Efficiency of Use:** Experienced users should be able to complete core tasks (e.g., creating a new project, adding a task) in less than 30 seconds.
- **User Satisfaction:** Users should find the interface intuitive and enjoyable to use.

6.2.2 Performance goals

The system must be responsive and handle concurrent users efficiently.

- **Response Time:** 90% of user interactions (e.g., loading a page, submitting a form) should complete within 2 seconds under normal load.
- **Concurrency:** The system should support 500 concurrent users without significant performance degradation.
- **Scalability:** The system should be able to scale to 5000 concurrent users with a proportional increase in resources, maintaining a 3-second response time for 90% of interactions.

6.2.3 Reliability goals

The product must be available and stable, minimizing downtime and data loss.

- **Availability:** The system should have an uptime of 99.9% annually, excluding scheduled maintenance.
- **Mean Time Between Failures (MTBF):** The average time between system failures should be at least 30 days.
- **Mean Time to Recover (MTTR):** The average time to restore full service after a failure should be no more than 4 hours.
- **Data Integrity:** All user data must be consistently stored and retrieved without corruption or loss.

6.2.4 Security goals

The product must protect user data and prevent unauthorized access.

- **Authentication:** All user access must be authenticated and authorized. Strong password policies (minimum 12 characters, including uppercase, lowercase, numbers, and symbols) and multi-factor authentication (MFA) will be enforced.

- **Authorization:** Users should only have access to data and functionalities for which they have explicit permissions (Role-Based Access Control).
- **Data Confidentiality:** All sensitive user data (e.g., personal information, project details) must be encrypted both in transit (SSL/TLS) and at rest (AES-256).
- **Vulnerability Management:** The system should be free from known critical and high-severity vulnerabilities.

7. Risk management

7.1 Risk list

List the identified risks or summary at the start of project. A separate and updated risk list is maintained at MMT.

Table 7.1. Project risks.

Risk ID	Explanation, severity/impact, probability, size/importance
1	Scope Creep: Medium impact (delays, resource strain), Medium probability (stakeholder requests, vague requirements), Medium importance (affects project timeline and quality).
2	Resource Unavailability: High impact (project delays, missed deadlines), Medium probability (key personnel leaving, unexpected absences), Large importance (critical path tasks dependent on specific resources).
3	Technical Challenges: Medium impact (development delays, quality issues), Medium probability (new technologies, complex integrations), Medium importance (affects product functionality and reliability).
4	Lack of Time Management: High impact (missed deadlines, project delays, rework), High probability (all group members working on other tasks), Large importance (affects overall project schedule and team morale).
5	Regulatory Compliance Issues: High impact (legal penalties, project halting), Low probability (thorough planning), Large importance (critical for project legality and public trust).

7.2 Risk monitoring

We'll be monitoring risks during weekly meetings (if responsible person is not named, then the task is up to the legendary Sam Body – nobody does it).

Risk management is done in the MMT tool.

How you plan to review/monitor the risks?

At least once at every sprint. Additionally, a dedicated risk review meeting will be held monthly, and urgent risks will be addressed as they arise through immediate communication with relevant stakeholders and a review of the MMT tool.

8 References

Will be added as we keep implementing stuff.

9 Open issues

No open issues for now.

10 Ideas for further development

If suunto watch integration is not completed until the end of this project we'll be at least giving a clean and proper integration roadmap that can be used by future team to implement that integration. Should the Suunto watch integration not be completed by the project's end, we will at minimum provide a clear and comprehensive integration roadmap. This roadmap can then be utilized by a future team to implement the integration effectively.

APPENDIX A

[Lumisovellus - current state](#)

APPENDIX B

[Lumisovellus - North Star Student Handout](#)